

Detailed Sedimentological Studies of the Glacigenic Sediments of Schirmacher Oasis, Antarctica

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Abstract

The present work emphasizes on the study of on going sedimentological processes in Schirmacher area as well as the detailed study of product of the same i.e., glacial debris and sediments. Physiographically, the Schirmacher area is divisible into continental ice, main Schirmacher land and shelf area. A clear distinction of the dominating physical agencies and their work in above mentioned three areas can easily be noticed. However, in a limited period of time, the present emphasis is given on various statistical analysis of the sediment samples collected from various sub-environments of the glacial environment. The common parameters like graphic mean, graphic skewness, graphic kurtosis, graphic standard deviation etc. have been calculated and interpreted for environmental reconstruction. In addition, a preliminary report of heavy minerals identified is also presented.

Keywords: Antarctica, Schirmacher range, Grain size analysis, Heavy minerals

Introduction

The report is an outcome of the author's visit to Antarctica as a summer team member in 21st expedition, 2002. The visit was aimed for the detailed sedimentological studies of the glacial sediments exposed in and around the Schirmacher range. The Schirmacher Range, a rock oasis is represented by undulating topography consisting of low elevations hills of 50-200mts. altitudes and depressions formed due to glacial valleys and lakes. It covers an area of 35 sq. kms. between Latitudes 77° 44' 30" - 77° 46' 30" S and 11° 22' 40" - 11° 54' 00"E Longitudes. While collecting the data during the fieldwork, it has been observed that there is an ample scope for the sedimentological studies. However, in the preliminary phase, an emphasis has been given on granulometric analysis of the sediments from different areas. Lai (1986) has already made a preliminary effort for heavy mineral analysis while Asthana and

Chaturvedi (1998) carried out their studies on the textural characteristics of the sediments and quartz grain morphology including SEM studies. Khun et al., (1993) have made a detailed study of the glaciomarine sediments of Weddell and Lazarev seas including Schirmacher oasis and interpreted about the glacial transport mechanism as well as the depositional process.

In the present study, an emphasis has been given on detailed granulometric analysis of the extensive sediment samples collected from different locations of Polar ice-sheet, Schirmacher main land including lakes and shelf area. Various statistical parameters have been calculated and interpreted for their significance in reconstruction of depositional environment. The heavy minerals from the same samples have also been separated and are under study, therefore, a preliminary report is being made.

Field Studies and Sampling

A total of thirty-seven sediment samples (A1-A28, A30 and A32-A39) have been collected from entire area. Basically, three major geomorphological units are clearly distinguishable in the field i.e., Polar ice-sheet, Schirmacher mainland including lakes and shelf area (Fig. 1). The first unit i.e., Polar ice-sheet covering a large area in the south is impregnated with abundant sand and silt size sediments. The sampling has been made on the surface, margins as well as at the base where sediments fall down with melt water and form debris. In some of the cases, small ice-chunks have been taken and grains were separated through melting followed by sieving. Eight samples (A2-A6 and A32-A34) have been collected from this area covering along the scarp face of ice-sheet exposed in south of Maitri, SASE point, Shivaling and NOVA side.

Adjacent to the Polar ice sheet, the second geomorphological unit is Schirmacher mainland is represented by Precambrian metamorphosed terrain consisting dominantly of high grade gneisses, crystalline schists, granite, charnokite etc. The terrain shows various geomorphological units as a result of glacial environments like - melt water channels, sub-glacial lakes, moraines, glacial tills etc. This range provides sufficient scope for the study of ongoing glacial process along with the glacial sediments from different sub-environments. However, the sampling in this area have been made with only two criteria in consideration i.e., lakes and adjoining area and second, rocky ground where influence of wind is a significant factor for sediment accumulation. Nine lake samples are collected from Priyadarshini lake; three other lakes located in south and west of Maitri,

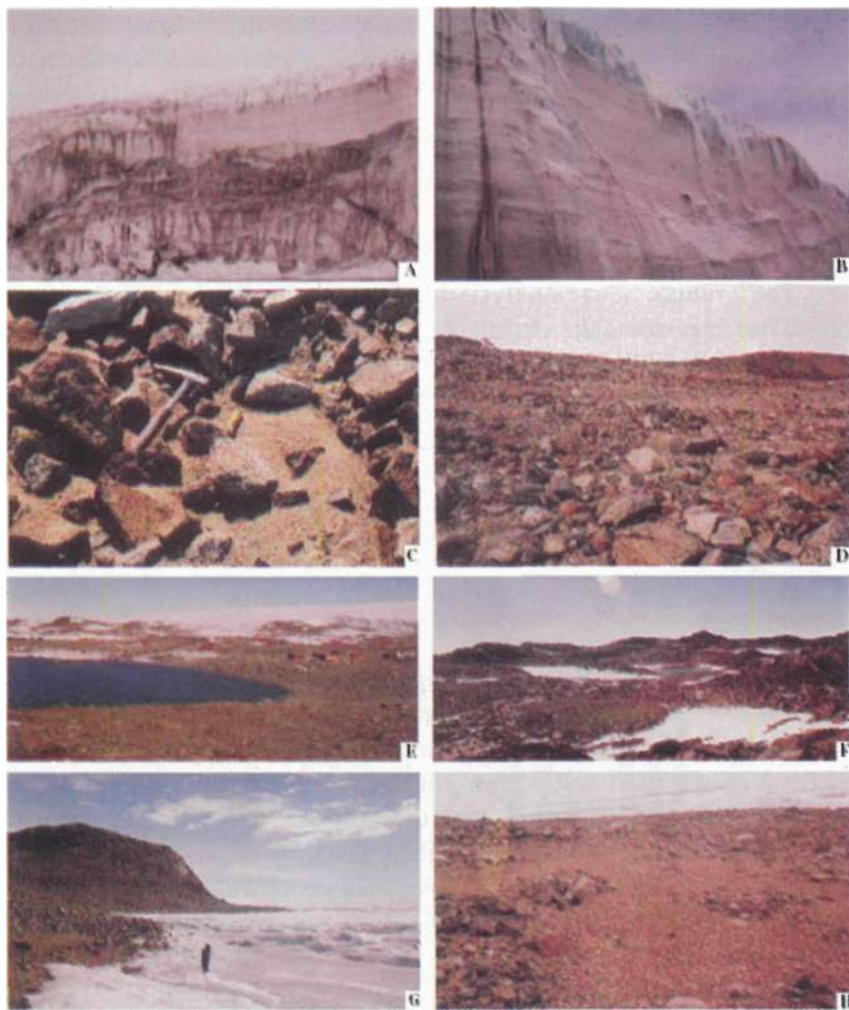


Fig. 1: A & B -Polar ice sheet exposed back side of the Maitri station; C & D -Sampling site from the rocky terrain of the Schumacher main land; E- Priyadarshini Lake; F- Two ice filled lake and one sweet water small lake in centre; G- Shelf area towards NOVA side and H- Mixed grain size' sediment exposure on the shelf at about five meters above the present water surface of the ocean

and in the route of NOVA; one in the route to shelf area (A7, A12-A13, A18-A20, A22, A28 & A30). The other category represented by seven numbers of samples is from rocky main land of the Schirmacher - surrounding Maitri station, in route to NOVA and towards the shelf region.

The third unit is the shoreline and the shelf region. It is marked by the presence of the mixed sediments coming from the Polar ice sheet as

well as mainland of Schirmacher through melt water channel. A total of thirteen samples are collected from shelf and adjoining area approachable towards NOVA side as well as along western margin (A8-A10, A15-16, A23-25, A27, and A35-A38).

Granulometric Analysis

The granulometric analysis of all thirty-seven samples has been carried out to generate the statistical data for comparison and other relevant interpretations. The methodology adopted is the traditional sieving technique for determination of grains size. The weight percentage and cumulative weight percentage of all samples were calculated for graphical representations in the form of cumulative curves and histograms. With the help of these statistical representations, various graphic measures have been computed.

Methodology

The sieving technique as proposed by Ingram (1971) has been followed for grain size analysis. Initially, hundred grams of sample was taken and directly subjected for removal of carbonate matter by treating it with 10% dil. HCl and washing it with distilled water. Similarly, the organic matter was removed with the help of 6% hydrogen peroxide followed by washing it with the distilled water.

The completely washed and dried sediments was subjected to sieve analysis. The B.S.S. (British Sieve Shaker) was arranged on quarter phi (Ö) intervals starting from 8 mesh (-1.25 Ö), 12 mesh (-0.75 Ö), 16 mesh (-0.25 Ö), 22 mesh (0.35 Ö), 30 mesh (0.75 Ö), 44 mesh (1.50 Ö), 66 mesh (2.00 Ö), 85 mesh (2.60 Ö), 120 mesh (3.0 Ö), 170 mesh (3.50 Ö), 240 mesh (4.10 Ö), and -240 mesh (4.25 Ö) and shake for thirty minutes. The retained fraction on each sieve was weighed accurately.

Graphical representation

The sieve analysis data have been represented by various graphical and statistical methods as proposed by McBride (1971).

(i) Weight percentage frequency

The values obtained from sieve analysis of hundred grams sample is directly used for weight percentage frequency of the sediments from

i) Polar ice sheet including melt-water channel, ii) Lake and adjoining area, iii) Schirmacher main rocky land, and iv) shelf area have been represented by histograms (Table 1; Figs 2,4,6,8 & 9). Broadly, the histogram shows dominance of medium to fine grained sediments, however, coarser sediments are present in all the four regions but recorded in comparatively more in the lakes and adjoining area sediments.

(ii) Cumulative weight percentage frequency curve

It has been constructed on the basis of the data obtained in Table 1, i.e., and weight percentage frequency. The data are cumulated on arithmetic ordinate scale beginning from -1.25 phi and continued up to 4.25 phi. The

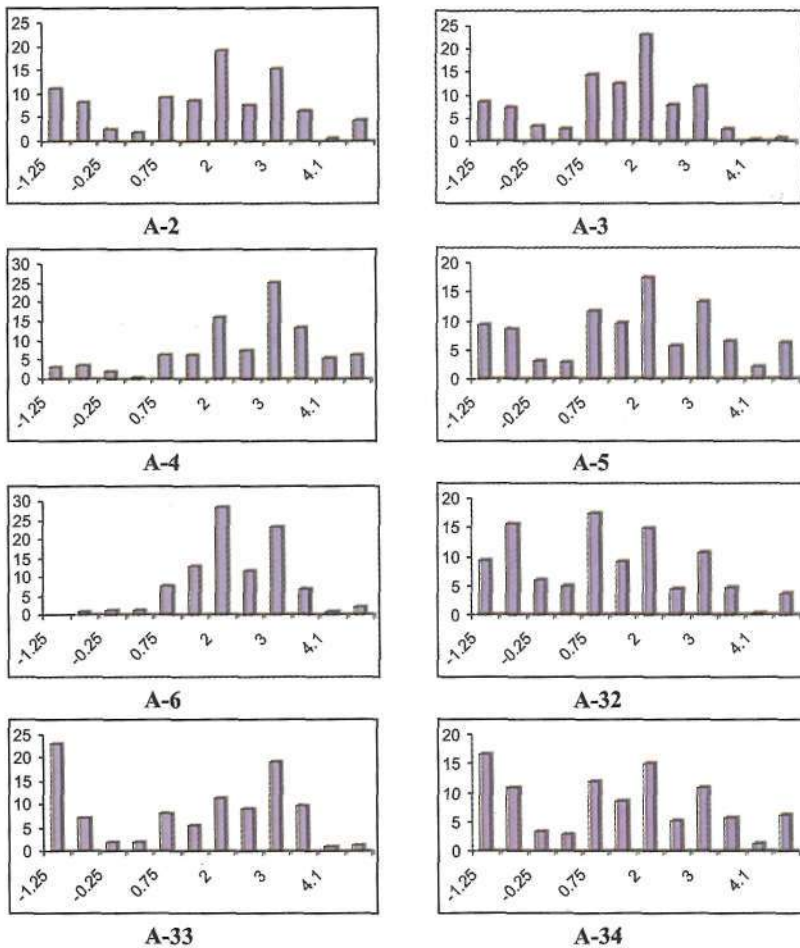


Fig. 2: Histograms of the samples from Polar ice-sheet and melt water channels

Table 1: Weight percent frequency in grams of the sediment samples from different area

Grain size Φ / Sample No.	-1.25	-0.75	-0.25	0.35	0.75	1.50	2.00	2.60	3.00	3.50	4.10	4.25
Polar ice-sheet and melt water channels												
A-2	11.31	8.40	2.57	1.79	9.32	8.75	19.38	7.74	15.38	6.27	0.66	4.38
A-3	8.62	7.32	3.19	2.65	14.63	12.39	23.02	7.74	11.89	2.73	0.29	0.72
A-4	3.27	3.55	2.12	0.57	6.22	6.34	16.07	7.40	25.34	13.58	5.36	6.42
A-5	9.40	8.59	3.17	2.73	11.63	9.74	17.35	5.77	13.21	6.57	2.18	6.16
A-6	0.18	0.77	1.01	1.01	7.56	12.78	28.45	11.56	23.40	6.80	0.63	1.98
A-32	9.39	15.63	6.05	4.88	17.44	9.16	14.75	4.47	10.64	4.71	0.38	3.65
A-33	23.11	7.16	1.91	1.96	8.21	5.48	11.23	9.01	19.15	9.76	1.08	1.22
A-34	16.74	10.84	3.32	2.76	12.04	8.57	15.07	5.16	11.02	5.74	1.33	6.31
Lake and adjoining area sediments												
A-7	8.58	10.34	5.37	3.47	13.26	8.71	15.83	5.95	13.83	6.37	1.60	3.48
A-12	3.22	13.91	6.34	3.99	12.30	6.38	10.48	4.56	11.78	6.96	4.10	6.94
A-13	21.83	15.68	5.27	4.78	18.10	7.79	9.04	2.78	5.99	4.24	1.34	1.90
A-18	6.36	8.16	5.06	3.41	17.85	12.97	18.49	5.31	9.05	4.10	2.47	2.58
A-19	16.89	7.36	2.78	2.05	8.38	6.68	12.97	5.74	14.88	11.72	4.81	4.59
A-20	19.99	6.58	2.28	1.61	7.34	6.19	14.17	6.76	16.28	11.00	3.59	3.19
A-22	13.56	8.65	3.36	2.35	10.64	8.36	14.59	6.40	14.72	10.25	3.88	2.58
A-28	8.26	6.43	3.27	2.96	14.49	11.73	20.15	7.32	14.77	7.10	0.31	3.37
A-30	24.23	7.66	3.15	2.23	7.57	5.27	9.84	5.44	15.73	11.51	3.20	2.63

(Contd.)

Table 1: Weight percent frequency in grams of the sediment samples from different area (Contd.)

Grain size Φ / Sample No.	-1.25	-0.75	-0.25	0.35	0.75	1.50	2.00	2.60	3.00	3.50	4.10	4.25
Schirmacher main rocky land												
A-1	0.00	0.20	0.48	0.79	9.47	12.35	0.0	35.04	17.31	10.70	0.21	10.88
A-11	14.83	8.31	3.28	2.92	15.83	12.26	15.51	5.44	9.78	5.68	1.20	2.39
A-14	2.77	1.77	0.83	0.71	5.08	8.28	23.49	11.53	25.18	10.97	3.50	4.81
A-17	6.65	2.96	0.95	0.73	5.47	7.72	21.83	10.01	23.67	10.84	3.12	4.95
A-21	2.67	1.60	1.51	1.92	20.26	22.06	29.45	7.02	10.29	2.78	0.41	0.47
A-26	7.36	6.85	2.72	1.67	5.84	4.57	11.73	6.29	23.05	15.56	5.23	8.81
A-39	16.13	7.95	3.65	5.11	11.57	8.68	15.71	6.89	12.92	6.84	0.74	0.61
Shelf area												
A-8	7.10	5.21	2.21	1.80	11.41	11.28	23.70	8.53	17.04	5.39	0.66	2.87
A-9	9.06	3.98	1.56	1.32	16.24	6.40	19.36	7.80	18.13	8.53	1.26	5.44
A-10	0.57	0.97	0.48	0.52	6.65	14.77	38.78	12.75	19.39	4.46	0.11	0.97
A-15	13.05	7.72	3.97	2.22	11.72	8.31	13.96	5.18	13.29	6.53	2.02	5.70
A-16	6.20	5.14	1.33	1.17	7.12	7.43	18.70	9.09	21.97	11.69	1.96	5.92
A-23	6.76	6.51	2.29	1.57	8.51	7.14	16.30	7.13	19.16	10.34	2.87	11.22
A-24	16.46	6.44	2.42	1.31	7.51	7.27	20.39	9.37	20.34	6.32	1.30	3.54
A-25	1.93	1.21	0.53	0.37	20.41	5.08	22.70	14.63	34.72	10.49	2.30	2.56
A-27	13.17	7.93	2.42	2.23	9.11	6.50	14.32	7.52	20.06	10.49	1.16	3.37
A-35	11.95	5.71	10.31	10.66	7.58	6.12	11.42	7.70	17.00	8.33	0.79	0.91
A-36	12.76	7.82	4.87	4.49	17.41	9.63	14.50	5.62	11.09	5.71	1.47	2.82
A-37	7.98	11.45	3.79	2.59	14.57	12.39	22.24	7.56	11.43	4.06	0.48	1.70
A-38	10.70	6.59	2.79	2.05	7.36	5.27	10.05	4.81	19.51	15.91	8.26	6.26

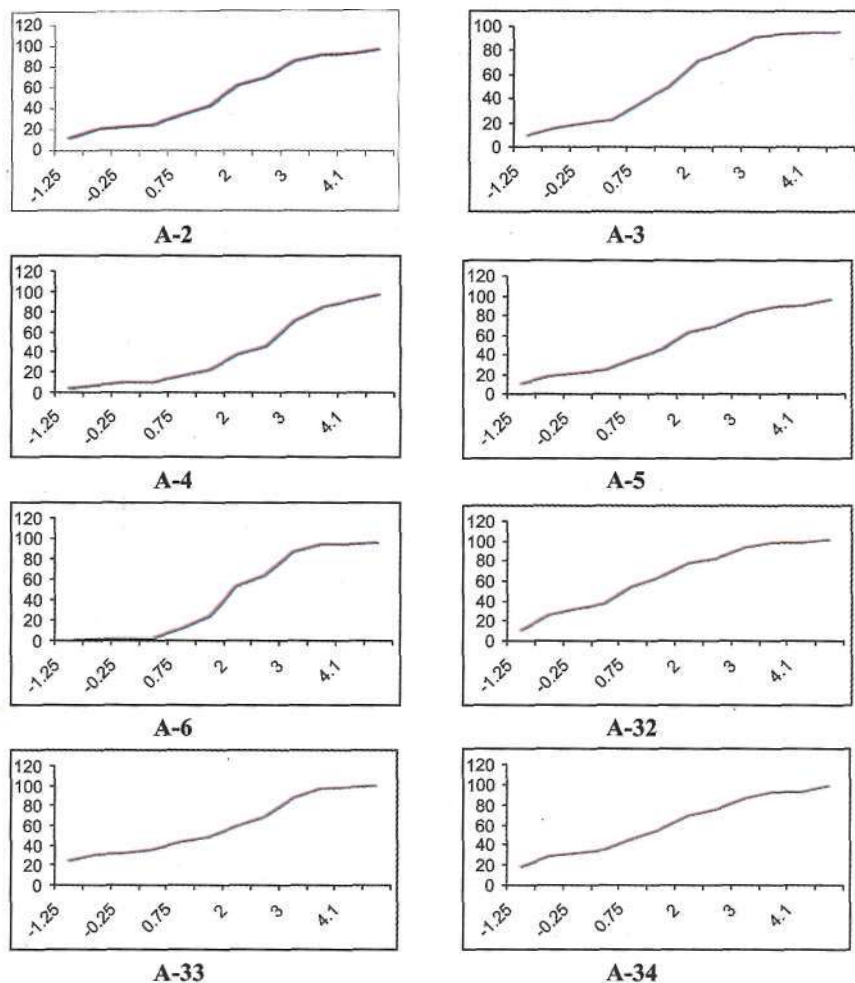


Fig. 3: Cumulative curves of the samples from Polar ice-sheet and melt water channels

graphical representation has been made by plotting grain size (in phi) on abscissa and cumulative weight percentage (in phi) on ordinate (Table 2; Figs 3,5,7,9 & 10). The detailed study shows the dominance of the medium to fine grained sediments in the samples of all the four areas, similar to the result carried out by weight percentage frequency.

Textural parameters

The cumulative frequency curve obtained by graphical representation served as the basis for all the determination carried out to know the various

Table 2: Cumulative weight percent frequency of the sediment samples from different area

Grain size Φ / Sample No.	-1.25	-0.75	-0.25	0.35	0.75	1.50	2.00	2.60	3.00	3.50	4.10	4.25
Polar ice-sheet and melt water channels												
A-2	11.31	19.71	22.28	24.07	33.39	42.14	61.52	69.26	84.64	90.91	91.57	95.95
A-3	8.62	15.85	19.04	21.69	36.62	48.71	71.73	79.47	91.39	93.99	94.28	95.00
A-4	3.27	6.83	8.94	9.51	15.73	22.07	38.14	45.54	70.88	84.46	89.82	96.24
A-5	9.40	17.99	21.16	23.89	35.52	45.26	62.61	68.38	81.59	88.16	90.34	96.50
A-6	0.18	0.95	1.96	2.97	10.53	23.31	51.76	63.32	86.72	93.53	94.15	96.13
A-32	9.39	25.02	31.07	35.95	53.39	62.55	77.30	81.77	92.41	97.12	97.50	101.15
A-33	23.11	30.27	32.18	34.14	42.35	47.83	59.06	68.07	87.22	96.98	98.06	99.28
A-34	16.74	27.58	30.90	33.66	45.70	54.26	69.33	74.49	85.51	91.25	92.58	98.89
Lake and adjoining area sediments												
A-7	8.58	18.92	24.29	27.76	41.02	49.73	65.56	71.51	85.34	91.71	93.31	96.79
A-12	3.22	17.13	23.47	27.46	39.76	46.14	56.62	61.18	72.96	79.92	84.02	90.96
A-13	21.83	37.51	42.78	47.56	65.66	73.45	82.49	85.27	91.26	95.50	96.84	98.74
A-18	6.63	14.52	19.57	22.98	40.83	53.80	72.29	77.60	86.65	90.75	93.22	95.80
A-19	16.89	24.26	27.04	29.10	38.48	44.17	57.14	62.89	77.77	89.50	93.68	98.27
A-20	19.95	26.56	28.86	30.48	37.82	44.02	58.19	64.96	81.24	92.25	95.44	98.64
A-22	13.56	22.21	25.57	27.92	38.56	46.92	61.51	67.91	92.63	92.88	96.76	99.34
A-28	8.26	14.70	17.97	20.94	35.43	47.17	67.32	74.65	89.42	96.53	96.84	100.22
A-30	24.23	31.89	34.04	37.27	44.84	50.11	59.95	65.63	81.12	92.63	95.83	98.46

(Contd.)

Table 2: Cumulative weight percent frequency of the sediment samples from different area (Contd.)

Grain size Φ / Sample No.	-1.25	-0.75	-0.25	0.35	0.75	1.50	2.00	2.60	3.00	3.50	4.10	4.25
Schirmacher main rocky land												
A-1	0.0	0.20	0.68	1.47	10.94	23.29	23.29	58.33	75.31	86.34	86.55	97.43
A-11	14.83	23.14	26.42	29.33	45.17	57.43	73.14	78.58	88.36	94.04	95.24	97.63
A-14	2.77	4.54	5.37	6.08	11.16	19.44	42.93	54.46	79.64	90.61	94.11	98.92
A-17	6.65	9.34	10.27	11.02	16.49	24.71	46.04	56.05	79.72	90.56	93.68	98.63
A-21	2.67	4.28	5.79	7.72	28.00	50.07	79.52	86.55	96.84	99.63	100.04	100.52
A-26	7.36	14.21	16.93	18.60	24.44	29.01	40.74	47.03	70.08	85.64	90.87	99.68
A-39	16.13	24.09	27.74	32.86	44.43	53.12	68.83	75.73	88.65	95.50	96.24	96.86
Shelf area												
A-8	7.10	12.11	14.32	16.12	27.53	38.81	62.51	71.04	88.08	93.47	94.13	97.00
A-9	9.06	13.04	14.60	15.92	32.16	38.56	57.92	65.72	83.85	92.32	93.64	99.08
A-10	0.57	1.54	2.02	2.54	9.19	23.96	62.74	75.49	94.88	99.34	99.45	100.42
A-15	13.05	20.77	24.75	26.96	38.68	46.99	60.95	66.13	79.42	85.95	88.03	93.73
A-16	6.20	11.34	12.67	13.84	20.97	28.40	47.10	56.19	78.16	89.86	91.82	97.74
A-23	6.76	13.27	15.56	17.13	25.64	32.78	49.08	56.21	75.37	85.71	88.58	99.80
A-24	16.46	22.90	25.32	26.63	34.15	41.43	61.81	71.18	91.52	97.85	99.15	102.69
A-25	1.93	3.14	3.67	4.04	6.45	11.53	34.23	48.86	83.58	94.07	96.37	98.93
A-27	13.17	21.10	23.52	25.75	34.86	41.36	55.68	63.20	83.26	93.74	94.90	98.27
A-35	12.95	18.66	28.98	39.64	47.23	53.35	64.78	72.48	89.49	97.82	98.62	99.53
A-36	12.76	20.58	25.45	29.94	47.35	56.98	71.48	77.10	88.19	93.90	95.37	98.19
A-37	7.98	19.43	23.22	25.81	40.38	52.77	75.01	82.57	94.00	98.06	98.54	100.24
A-38	10.70	17.29	20.08	22.13	19.49	34.76	44.81	49.62	69.13	85.04	93.30	99.56

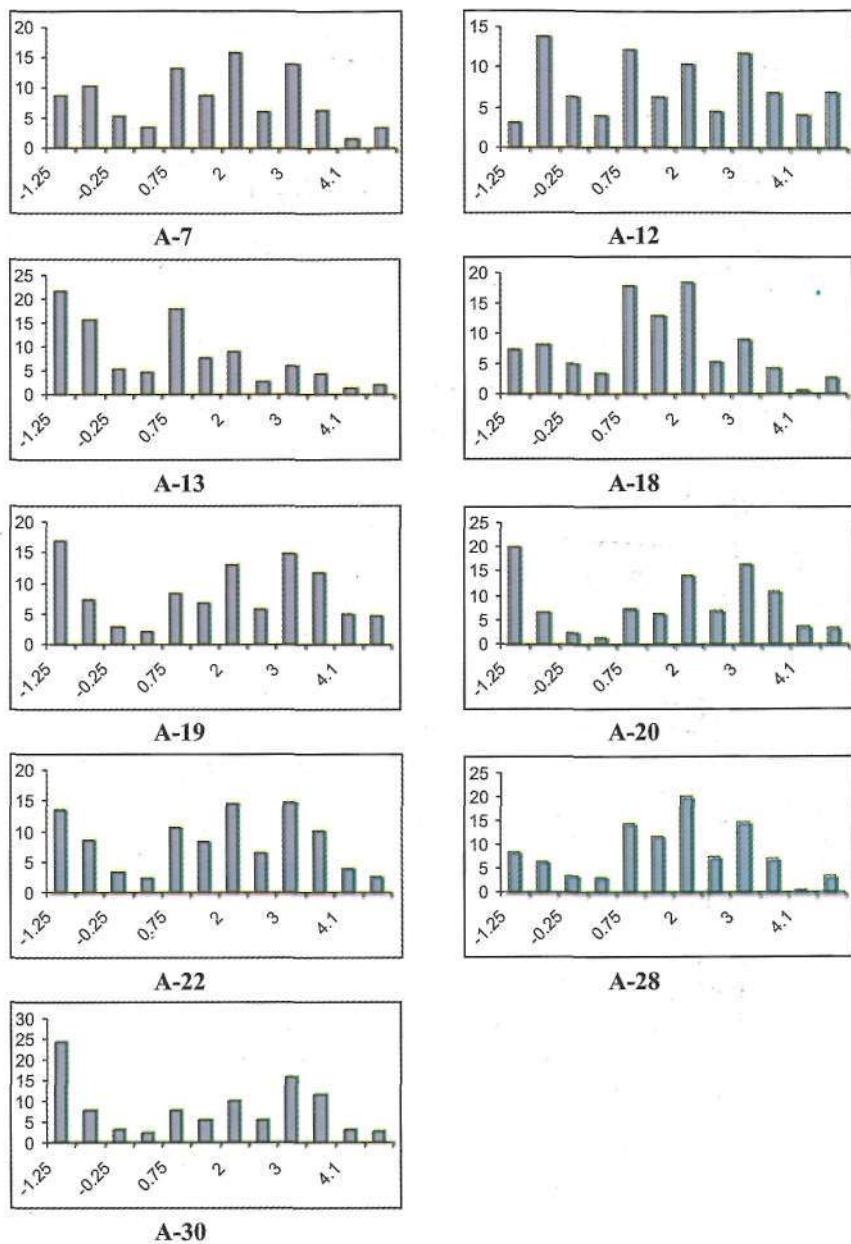


Fig. 4: Histograms of the samples from lake and adjoining area

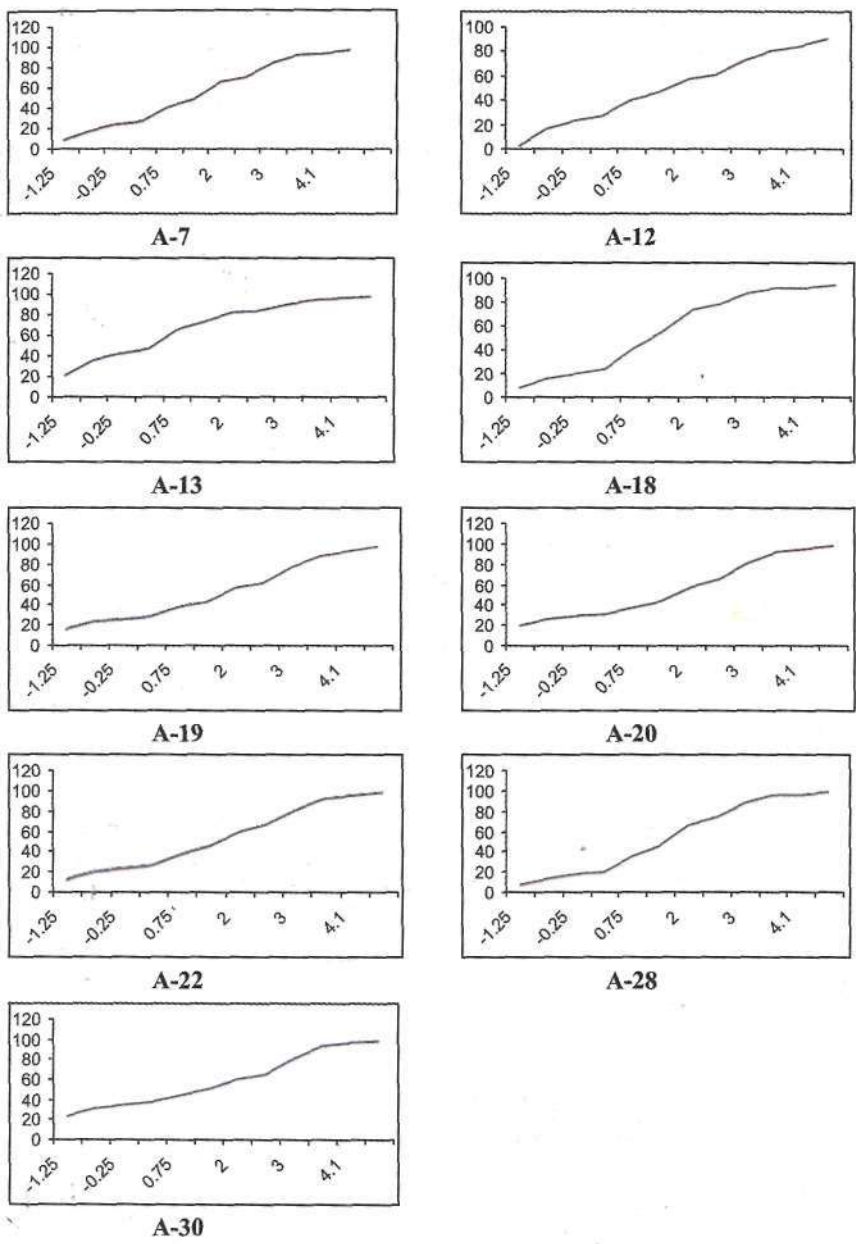
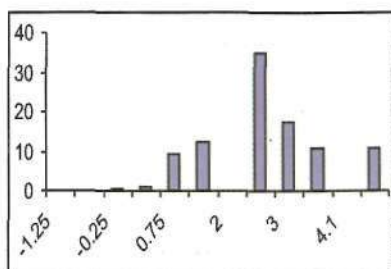
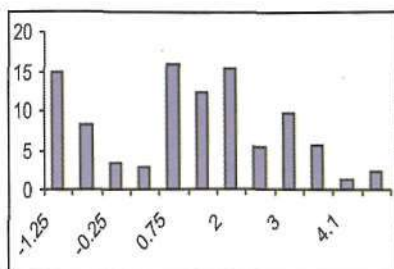
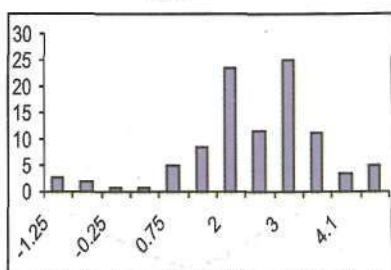
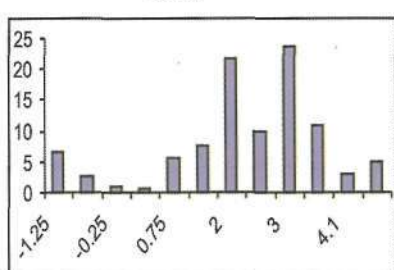
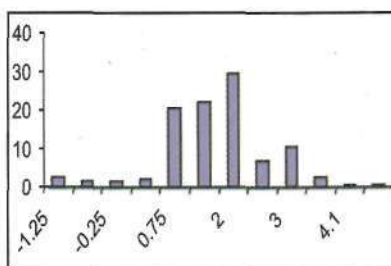
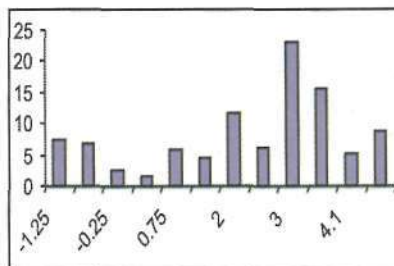
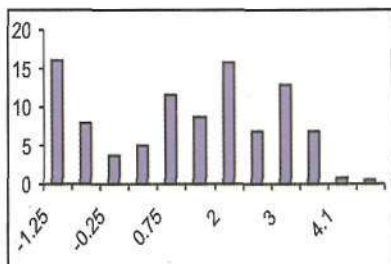


Fig. 5: Cumulative curves of the samples from lake and adjoining area

**A-1****A-11****A-14****A-17****A-21****A-26****A-39***Fig. 6: Histograms of the samples from Schirmacher main land*

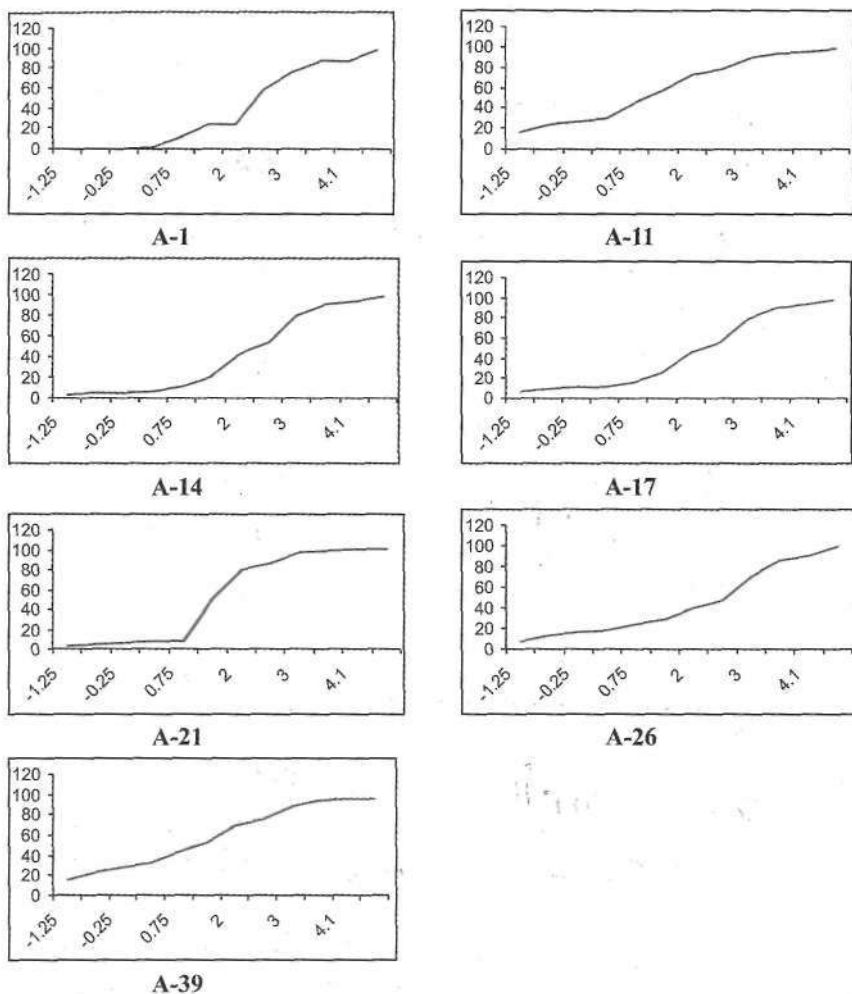


Fig. 7: Cumulative curves of the samples from Schirmacher main land

textural parameters like graphic mean, graphic median, graphic skewness, graphic kurtosis, graphic standard deviation etc. The methodology followed and interpretations are as proposed by Folk and Ward (1957). The details of the various parameters are as follows (Table 3).

(i) Graphic mean size:

The mean size denotes a measure of central tendency and is calculated with the help of the formula; $\text{Gr. mean} = \Phi_{16} + \Phi_{50} + \Phi_{84} / 3$,

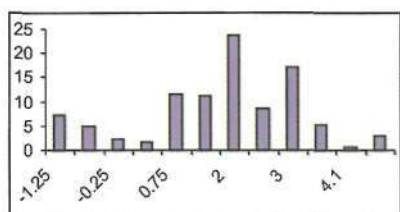
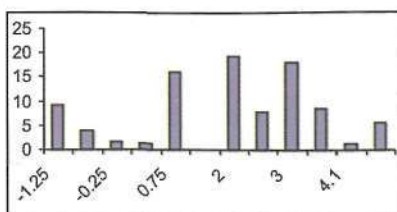
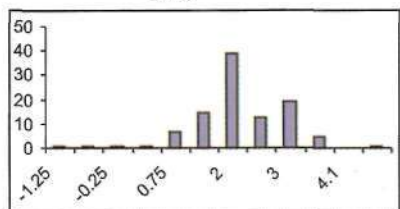
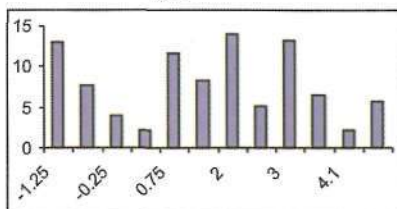
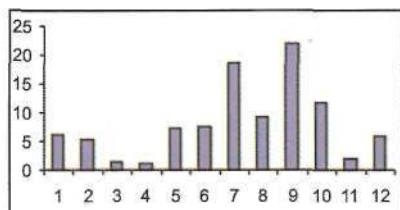
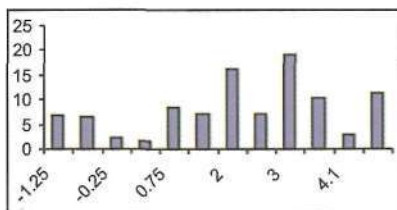
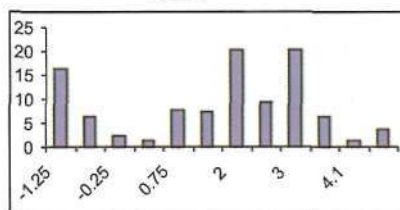
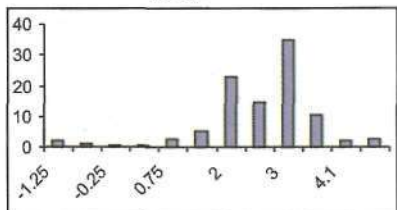
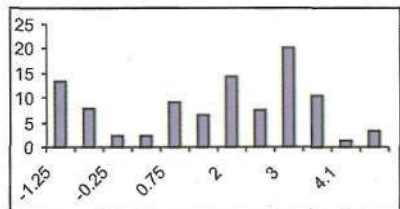
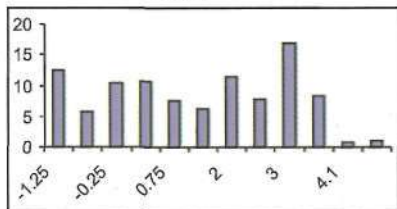
Table 3 : Graphic measures from the grain size analysis of the sediment samples from different area

Stat. Paramet./ Sample No.	Φ95	Φ84	Φ75	Φ50	Φ25	Φ16	Φ5	M_z	σ_1	Sk_1	K_G
Polar ice-sheet and melt water channels											
A-2	4.41	2.98	2.74	1.69	0.36	-0.94	-1.53	1.24	1.88	+0.21	1.10
A-3	4.45	2.76	2.26	1.51	0.45	-0.77	-1.40	1.17	1.78	-0.14	1.32
A-4	4.42	3.41	3.13	2.66	1.57	0.74	-1.04	2.27	1.48	+0.38	1.43
A-5	4.37	3.15	2.81	1.62	0.38	-0.89	-1.44	1.29	1.81	+0.14	0.98
A-6	4.32	2.95	2.81	1.97	1.52	1.06	0.47	1.99	1.00	+0.12	1.22
A-32	3.31	2.69	1.95	0.67	-0.73	-1.03	-1.37	0.77	1.63	+0.11	0.71
A-33	3.41	2.95	2.76	1.60	-1.14	-1.77	-2.55	0.98	2.08	0.40	0.62
A-34	4.24	2.96	2.61	1.12	-0.85	-1.29	-1.78	0.93	1.98	-0.05	0.71
Lake and adjoining area sediments											
A-7	4.25	2.94	2.71	1.49	-0.23	-0.89	-1.43	1.18	1.81	+0.13	0.73
A-12	4.71	4.05	3.15	1.67	0.0	-0.49	-1.17	1.64	2.13	-0.006	0.76
A-13	3.41	2.44	1.61	0.41	-1.14	-1.43	-1.82	0.47	1.75	+0.09	0.77
A-18	4.45	2.83	2.21	1.25	0.36	-0.64	-1.38	1.14	1.74	+0.01	1.29
A-19	4.20	3.29	2.93	1.73	-0.66	-1.31	-2.07	1.23	2.19	+0.26	0.71
A-20	4.02	3.15	2.84	1.72	-0.87	-1.53	-2.40	1.11	2.04	+0.33	0.70
A-22	3.89	3.08	2.80	1.61	-0.31	-1.09	-1.75	1.20	1.89	+0.24	0.74
A-28	3.41	2.85	2.59	1.58	0.48	-0.58	-1.45	1.28	1.50	+0.25	0.94
A-30	3.94	3.15	2.84	1.51	-1.17	-1.81	-2.57	0.95	2.22	+0.28	0.66

(Contd.)

Table 3 : Graphic measures from the grain size analysis of the sediment samples from different area (Contd.)

Stat. Paramet./ Sample No.	Φ95	Φ84	Φ75	Φ50	Φ25	Φ16	Φ5	M_z	σ_1	Sk_1	K_G
Schirmacher main rocky land											
A-1	4.39	3.38	3.10	2.46	2.02	1.06	0.49	2.3	1.17	+0.10	1.48
A-11	3.74	2.81	2.20	1.03	-0.49	-1.17	-1.72	0.89	2.25	+0.10	0.83
A-14	4.12	3.22	2.93	2.37	1.62	1.23	-0.79	2.27	1.23	+0.21	1.53
A-17	4.23	3.20	2.93	2.22	1.51	0.70	-1.68	2.07	1.52	+0.25	1.70
A-21	2.93	2.39	1.91	1.50	1.05	0.90	-0.58	1.59	0.89	-0.01	1.12
A-26	4.28	3.44	3.16	2.64	0.75	0.47	-1.38	2.18	1.59	+0.37	0.96
A-39	3.45	2.86	2.55	1.22	-0.58	-1.25	-1.96	0.94	1.83	+0.19	0.70
Shelf area											
A-8	4.22	2.92	2.70	1.73	0.66	0.19	-1.42	1.61	1.53	+0.07	1.13
A-9	4.60	3.39	2.95	1.95	0.56	0.31	-1.49	1.88	1.69	+0.15	1.03
A-10	3.09	2.77	2.56	1.85	1.51	1.09	0.47	1.90	0.81	+0.04	1.03
A-15	4.56	3.39	2.86	1.62	-0.26	-1.06	-1.78	1.31	2.08	-0.06	0.83
A-16	4.30	3.26	2.14	2.19	1.22	0.48	-1.36	1.97	1.55	+0.23	1.35
A-23	4.35	3.42	2.98	2.09	0.69	-0.16	-1.31	1.78	1.75	+0.22	1.05
A-24	3.76	2.87	2.69	1.72	-0.32	-1.26	-2.14	1.11	1.91	+0.37	2.02
A-25	3.71	3.05	2.90	2.62	1.82	1.61	0.54	2.42	0.84	-0.15	1.20
A-27	4.15	3.05	2.83	1.92	0.07	-1.05	-1.79	1.27	1.92	+0.30	0.88
A-35	3.35	2.86	2.68	1.02	-0.47	-0.97	-1.97	0.97	1.75	+0.08	0.69
A-36	4.05	2.85	2.42	0.93	-0.33	-1.05	-1.77	0.91	1.66	+0.03	0.86
A-37	3.14	2.66	2.00	1.33	0.26	-0.91	-1.37	1.02	1.57	+0.22	1.05
A-38	4.18	3.44	3.19	2.59	0.51	-0.83	-1.52	1.73	1.92	+0.52	0.87

**A-8****A-9****A-10****A-15****A-16****A-23****A-24****A-25****A-27****A-35****Fig. 8: Histograms of the samples from shelf area**

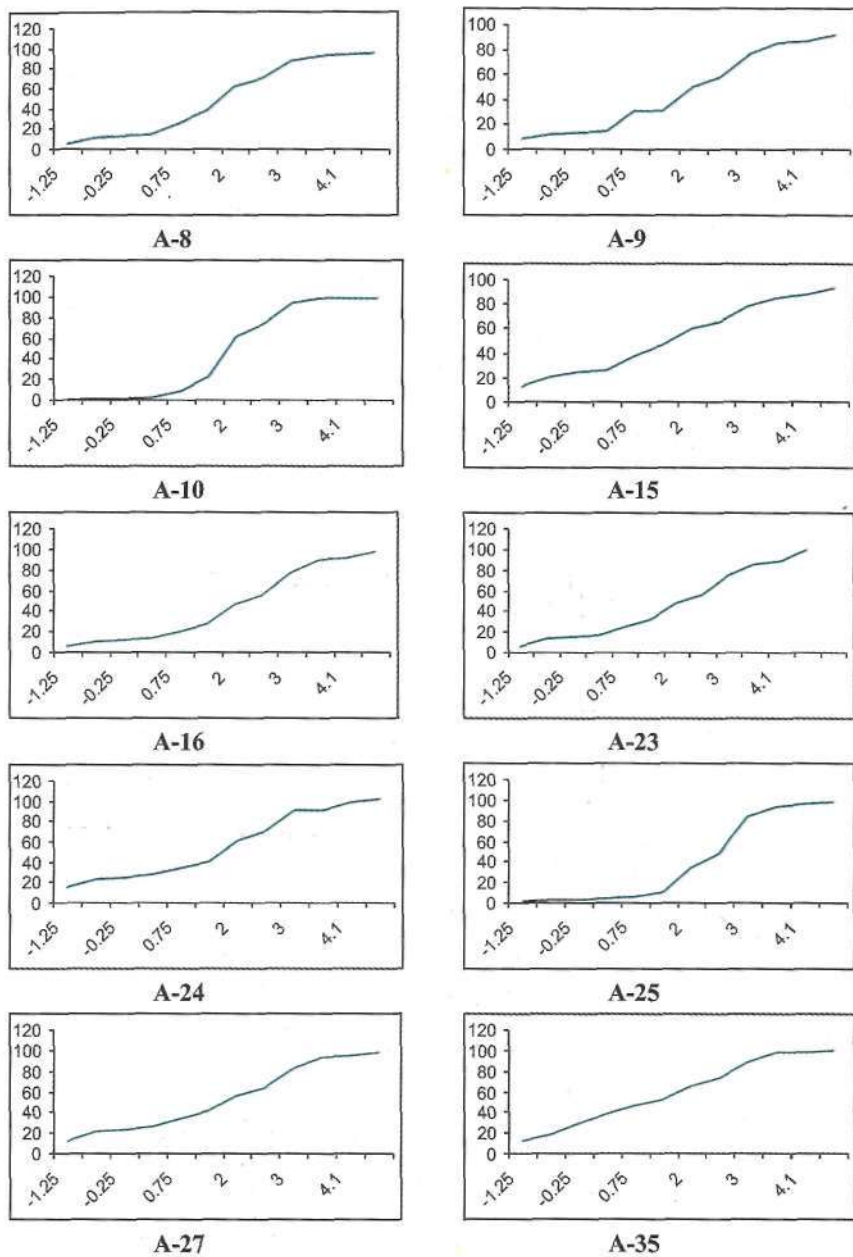


Fig. 9: Cumulative curves of the samples from shelf area

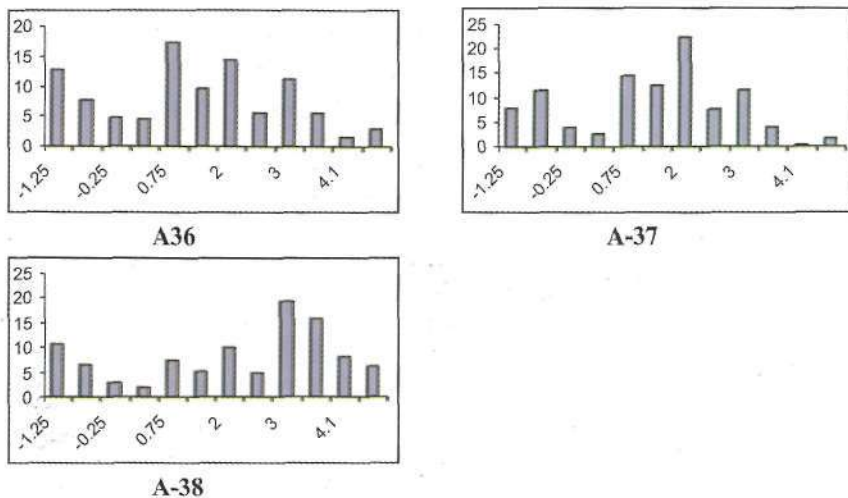


Fig.10: Histograms of the samples from shelf area.

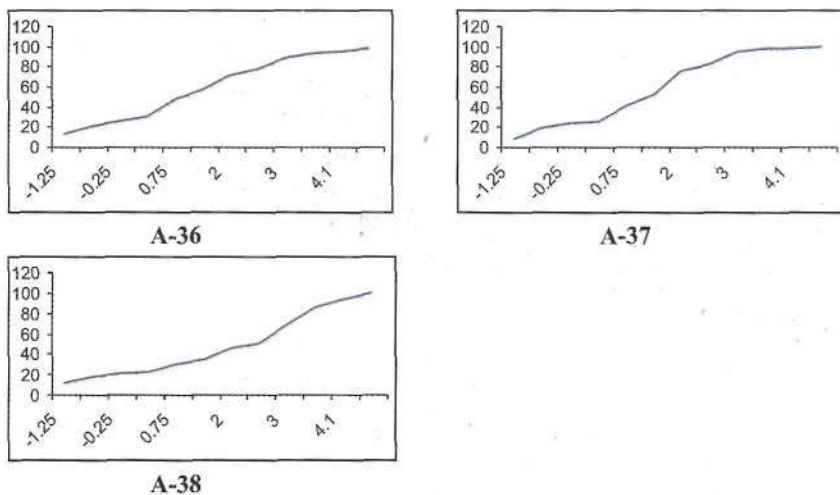


Fig. 11: Cumulative curves of the samples from shelf area

the graphic mean size of the samples from, i) Polar ice sheet including melt-water channel, ii) Lake and adjoining area, iii) Schirmacher main rocky land, and iv) Shelf area have been calculated which ranges from 0.77 to 2.27, 0.47 to 1.28, 0.94 to 2.30 and 0.81 to 2.08 respectively. In general, all the four area shows dominance of medium to fine grain sediments.

(ii) Graphic median:

It is $\ddot{O}_{50}$ value that denotes half of the particles (by weight) are coarser and half are fine in size. It is computed ranging from 0.67 to 2.66, 0.41 to 1.72, 1.22 to 2.46 and 0.93 to 2.59 respectively for the four area as mentioned in graphic mean size. The values again show general tendency of dominance of fine to medium grains.

(iii) Graphic standard deviation:

The graphic standard deviation measures the sorting or uniformity of particle size distribution. It is calculated by the formula; Gr. S.D = $\ddot{O}_{84} - \ddot{O}_{16} / 4 + \ddot{O}_{95} - \ddot{O}_5 / 6.6$. The values exhibits that the sediments from i) polar ice sheet and melt water channels are dominantly poorly sorted (1.00 to 2.08); ii) lake and adjoining area are poorly to very poorly sorted (1.50 to 2.22); iii) Schirmacher main land area show moderate to poor sorting (0.89 to 2.25) and iv) shelf area represents poor sorting (0.81 to 2.08).

(v) Graphic skewness:

Graphic skewness is calculated by the formula, Gr. S.K = $\ddot{O}_{84} + \ddot{O}_{16} - 2\ddot{O}_{50} / (\ddot{O}_{84} - \ddot{O}_{16}) + \ddot{O}_{95} + \ddot{O}_5 - 2\ddot{O}_{50} / (\ddot{O}_{95} - 2\ddot{O}_5)$. It measures the symmetrical distribution, i.e., predominance of coarse or fine sediments. If the coarse tail is more (coarse sediments) then it is considered negative skewness, while positive skewness when fine tail is more, i.e., predominance of fine matter.

The values calculated shows that the sediments from Polar region mostly fall in between nearly symmetrical to fine skewed category (-0.05 to +0.21) while two samples are finely skewed. The lake and adjoining area sediments are dominantly of two categories i.e., nearly symmetrical (-0.006 to +0.33). The Schirmacher main land shows a range from nearly symmetrical to very fine skewed (-0.01 to +0.37) whereas, the shelf area is again represented mainly by nearly symmetrical to fine skewed (-0.06 to +0.300, however, one sediment is finely skewed (+0.37)

(vi) Graphic Kurtosis:

The graphic kurtosis relates to the peakness of the distribution. It measures the ratio between the sorting in the tail as well as in the central portion. The condition when the central portion showing better sorting than the tail then it is said to be leptokurtic or extremely peakedness curve; where as the vice-a-versa condition is known as platkurtic or flat peaked.

The in between situation, i. e., both tail and peaked equally sorted condition is known as mesokurtic. It is calculated with the help of the formula; $Gr. Kurt. = \ddot{O}_{95} - \ddot{O}_5 / 2.44 (\ddot{O}_{75} - 2\ddot{O}_{25})$.

The graphic kurtosis values shows dominance of platykurtic to leptokurtic sediments in Polar ice sheet and melt water channel region (0.71 to 1.43) whereas the lake and adjoining area sediments are mostly platykurtic (0.66 to 0.77). Schirmacher main land as well as shelf area both show platykurtic to leptokurtic sediments with a different value range i.e., 0.70 to 1.53 and 0.83 to 2.02 respectively.

The graphic mean size analysis indicates a medium to fine range of grains. The sorting is poor to very poor. The sediments are nearly symmetrical to finely skewed and platykurtic to leptokurtic in nature. The wide range of grain size classes, poorly sorted nature, nearly symmetrical to fine skewed and platykurtic to leptokurtic grain size are the typical characteristics of glacial deposits (Friedman, 1962; Reinick and Singh, 1980; Sengupta, 1996).

Heavy Mineral Analysis

All the thirty-seven samples used for grain size analysis have been taken for the heavy mineral analysis. The gravity settling separation or funnel separation method is used as proposed by Carver (1971) and Lindholm (1987). Initially, ten grams of the 100 mesh (150 micron) sample prepared for granulometric analysis have been taken and settled in bromoform. The settled heavy crop is then washed, dried and weighted followed by mounting it on the glass slides for microscopic observations.

The heavy assemblage is represented by zircon, tourmaline, rutile, garnet, epidote hornblende and opaque minerals. Zircon is pink and colorless, anhedral to subhedral and subangular to subrounded shape showing high order interference color and high relief; with beautiful idiomorphic outlines. It is frequently observed in majority of the samples. Tourmaline is prismatic, angular to sub-angular in shape showing brown to deep brown pleochroism in cross nicol. It is also very frequently observed grain in most of the samples. Rutile is anhedral to sub-rounded with cross fractures, colorless with yellowish and reddish tint having thick black border showing high order interference color under cross nicols. Garnet showing joints and fractures are subangular to subrounded and isotropic between cross nicols. Epidote shows pink to pistachio green color, subrounded to rounded shape, high refractive index with anomalous interference color whereas amphibole represented by only hornblende is dark green color, prismatic, elongated and subhedral shape showing

strongly pleochroic shades of green. The opaque minerals are mainly iron oxide and are represented by magnetite, hematite and ilimanite. They are dark brown to black in color and occur as subrounded to rounded grains.

The heavy crop represents a mixed assemblage i.e., most stable/ultra stable suit represented by zircon, tourmaline and rutile followed garnet and epidote belonging to metastable and lastly unstable mineral i.e., hornblende. The categorization is based on the criteria proposed by Folk (1961) and Pettijohn, et al., (1973). The assemblage apparently dominated by tourmaline, garnet, rutile and zircon indicates the derivations from the metamorphic source (Force, 1980; Borghain et al., 1999). The work on other various interpretations is still in progress.

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